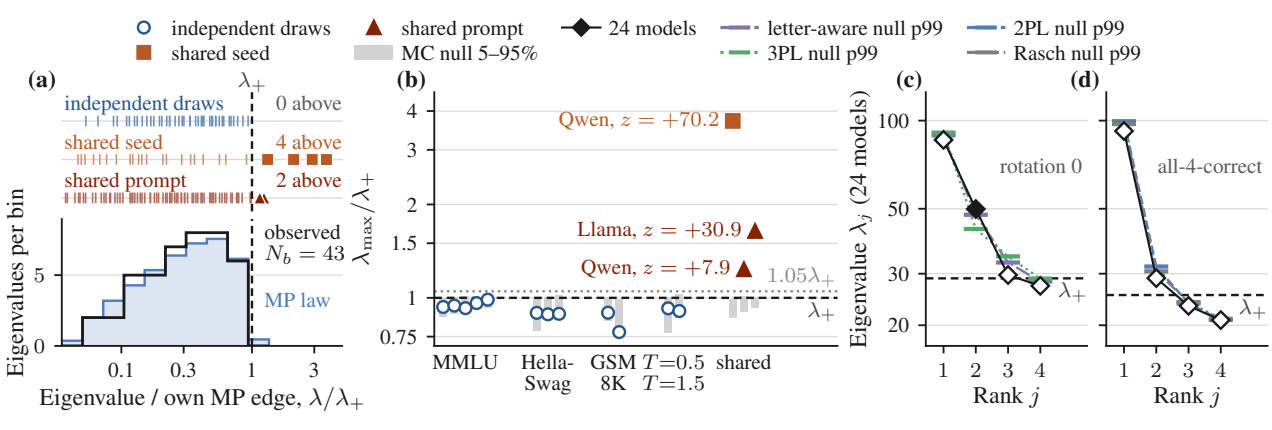

independent draws
shared seed
shared prompt
MC null 5–95%
24 models
letter-aware null p99
3PL null p99
2PL null p99
Rasch null p99
(a)
λ_+
independent draws
0 above
shared seed
4 above
shared prompt
2 above
observed
$N_b = 43$
MP law
Eigenvalues per bin
Eigenvalue / own MP edge, λ/λ_+
(b)
$\lambda_{\max}/\lambda_+$
Qwen, $z = +70.2$
Llama, $z = +30.9$
Qwen, $z = +7.9$
$1.05\lambda_+$
λ_+
MMLU
Hella-Swag
GSM 8K
T=0.5
T=1.5
shared
(c)
(d)
Eigenvalue λ_j (24 models)
rotation 0
all-4-correct
Rank j